LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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TITLE

Design and Implementation of a Pure Semantic HTML Static Webpage Without CSS

AIM

To develop a pure semantic HTML5 static webpage for a fictitious site “StaticWeb” using structural elements such as header, nav, section, main, article, aside, and footer, combined with table-based layout regions and native HTML form controls, entirely without CSS styling.

OBJECTIVE

- To practice document structuring using semantic HTML5 elements.
- To use structural tags (header, nav, main, article, aside, footer, section) to organize page content.
- To build a two-column page layout using an HTML table for the main content and sidebar regions.
- To construct ordered and unordered lists for sidebar reference content.
- To implement a native HTML contact form with text, email, select, and textarea inputs inside a fieldset.

THEORY

HTML5 introduces semantic elements that describe the meaning of page regions rather than their appearance. In the absence of any CSS, browsers apply sensible default styling to these elements, while screen readers and web scrapers can still interpret the page structure correctly from tags such as header, nav, main, article, section, aside, and footer. Combining these semantic wrappers with a structural table for column layout and native form controls demonstrates how far raw HTML can go in producing an organized, accessible, and functional static page.

Application Flow:

HTML5 Structure → Header & Navigation → Hero Section → Article Content & Sidebar → Contact Form → Footer

ALGORITHM / PROCEDURE

1. Create an HTML5 document with the title ‘Pure Semantic HTML Static Webpage’.
2. Build a header section containing the site title “StaticWeb” and a navigation bar, laid out using an HTML table.
3. Add a hero/showcase section introducing the page content, centered using the <center> tag.
4. Create a structural two-column table: a 70% wide main column and a 30% wide shaded sidebar column.

5. Inside the main column, add a <main> element containing two <article> blocks describing core structural tags and native interaction elements.
6. Inside the second article, implement a contact form using <fieldset> and <legend>, with text, email, select, and textarea inputs.
7. Inside the sidebar column, add an <aside> containing an unordered list of benefits and an ordered list of required components.
8. Add a footer section with a copyright notice.
9. Verify the webpage rendering and form controls in a browser.

PROGRAM

index.html

```
week1.html > html > body > footer > center
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Pure Semantic HTML Static Webpage</title>
</head>
<body>

    <!-- Header Section -->
    <header>
        <table width="100%" border="0" cellspacing="0" cellpadding="10">
            <tr>
                <td>
                    <h1>StaticWeb</h1>
                </td>
                <td align="right">
                    <nav>
                        <a href="#home">Home</a>  | 
                        <a href="#articles">Articles</a>  | 
                        <a href="#about">About</a>  | 
                        <a href="#contact">Contact Us</a>
                    </nav>
                </td>
            </tr>
        </table>
    </header>

    <hr>

    <!-- Hero / Showcase Section -->
    <section id="home">
        <center>
            <h2>Semantic Structures Without CSS</h2>
            <p>This layout uses raw HTML components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.</p>
            <p><a href="#articles"><b>Read Our Articles Below &darr;</b></a></p>
        </center>
    </section>

    <hr>

    <!-- Main Content Layout using a Structural Table -->
    <table width="100%" border="0" cellspacing="0" cellpadding="20">
        <tr valign="top">

            <!-- Left Side: Main Content Column -->
            <td width="70%">
                <main id="articles">

                    <article>
                        <h2>1. Core Structural Tags</h2>
```

```html
<article>
    <h2>1. Core Structural Tags</h2>
    <p>Published: July 7, 2026</p>
    <p>When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like <code>&lt;main&gt;</code>, <code>&lt;article&gt;</code
    <p>Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.</p>
</article>

<br><hr><br>

<article>
    <h2>2. Native Interaction Elements</h2>
    <p>Published: July 5, 2026</p>
    <p>Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to back

    <h3>Interactive Demonstration Forms</h3>
    <form id="contact" action="#" method="post">
        <fieldset>
            <legend><b>User Inquiry Form</b></legend>
            <p>
                <label for="name">Full Name: </label>
                <input type="text" id="name" name="username" required placeholder="John Doe">
            </p>
            <p>
                <label for="email">Email Address: </label>
                <input type="email" id="email" name="useremail" required>
            </p>
            <p>
                <label for="topic">Inquiry Topic: </label>
                <select id="topic" name="usertopic">
                    <option value="general">General Feedback</option>
                    <option value="technical">Technical Architecture</option>
                    <option value="academics">Academic Curriculum</option>
                </select>
            </p>
            <p>
                <label for="message">Message Details:</label><br>
                <textarea id="message" name="usermessage" rows="5" cols="40" placeholder="Type your notes here..."></textarea>
            </p>
            <p>
                <input type="submit" value="Submit Form Data">
                <input type="reset" value="Clear Entries">
            </p>
        </fieldset>
```

```html
                            <p>
                                <input type="submit" value="Submit Form Data">
                                <input type="reset" value="Clear Entries">
                            </p>
                        </fieldset>
                    </form>
                </article>

            </main>
        </td>

        <!-- Right Side: Sidebar Column -->
        <td width="30%" bgcolor="#f4f4f4">
            <aside id="about">
                <h3>Document References</h3>
                <p>Static pages map data out directly without real-time database queries or processor overhead.</p>

                <h4>Core Benefits:</h4>
                <ul>
                    <li>Instant browser painting</li>
                    <li>Low memory footprints</li>
                    <li>High data compatibility</li>
                </ul>

                <h4>Required Components:</h4>
                <ol>
                    <li>Document Type Definition</li>
                    <li>Semantic Wrapper Blocks</li>
                    <li>Data Input Control Interfaces</li>
                </ol>
            </aside>
        </td>

    </tr>
</table>

<hr>

<!-- Footer Section -->
<footer>
    <center>
        <p>&copy; 2026 Pure HTML Architecture. Assembled completely without style configurations.</p>
    </center>
</footer>
```

CODE EXPLANATION

Code / Syntax	Explanation
<header>, <nav>	Groups the site title and navigation links at the top of the page inside a semantic header region.
<table> (layout)	A non-semantic table used purely to arrange the header row and the two-column body layout, since no CSS is used for positioning.
<section>, <article>	Marks the hero introduction and each independent piece of article content so its structure is clear without any styling.
<main>	Wraps the primary page content (the articles), distinguishing it from header, sidebar and footer regions.
<aside>	Identifies the sidebar column as content related to, but separate from, the main article content.
<fieldset>, <legend>	Groups the related form controls of the inquiry form and gives the group a visible caption.
<input>, <select>, <textarea>	Native form controls used to capture the name, email, inquiry topic, and free-text message.
<footer>	Contains the closing copyright notice for the page.

TEST CASES AND EXECUTION DATA

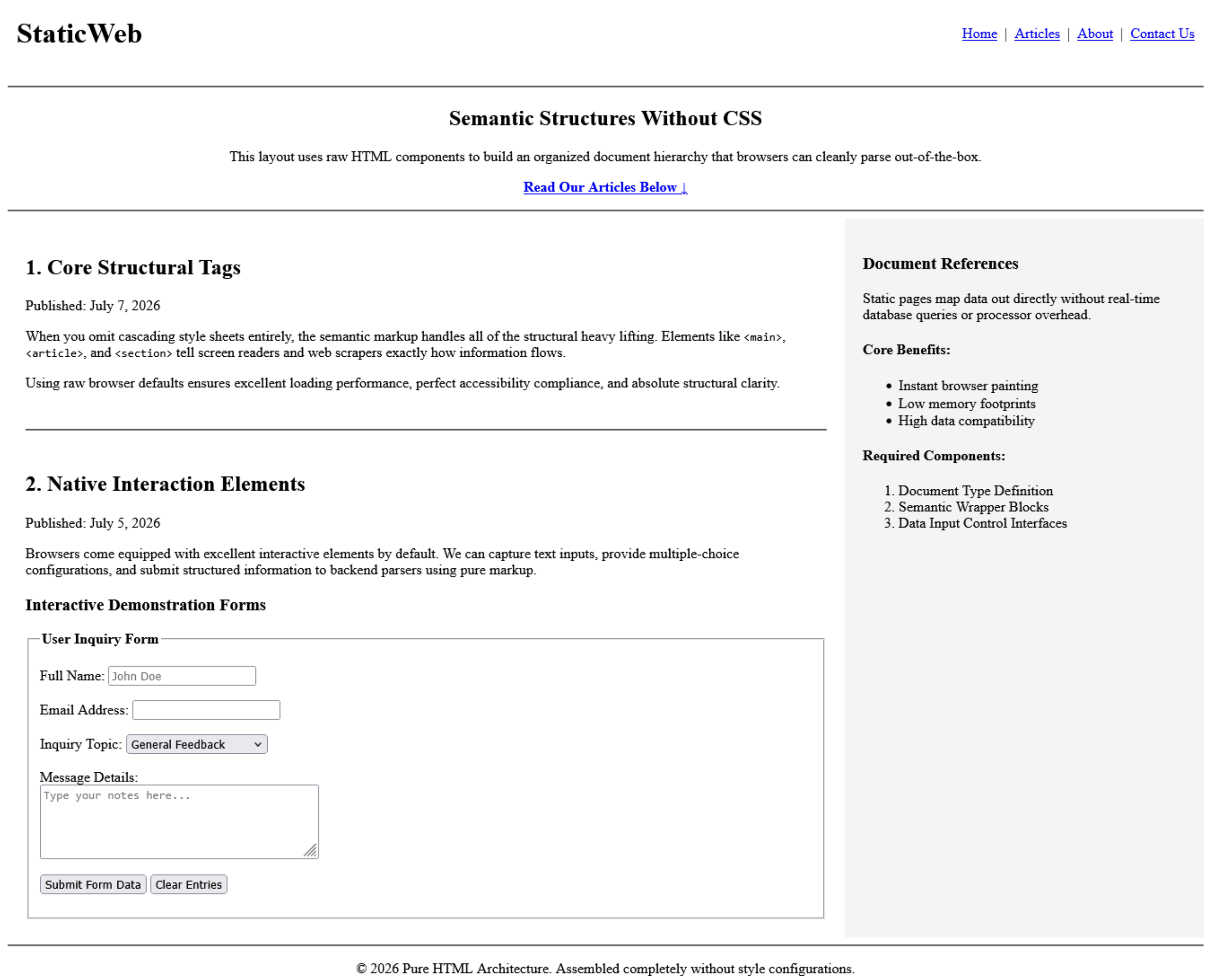
The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Header & Navigation	Inspect site title and nav links	"StaticWeb" title and four nav links render correctly, each jumping to its section	Pass
TC-02	Hero Section	Inspect centered heading and intro text	Heading, paragraph, and link appear centered on the page	Pass
TC-03	Two-Column Layout	Inspect main content vs. sidebar	70% main column and 30% shaded sidebar column render side by side	Pass
TC-04	Sidebar Lists	Check unordered and ordered lists	"Core Benefits" shown as bullets; "Required Components" shown as a numbered list	Pass

TC-05	Contact Form Fields	Fill name, email, topic dropdown, and message	Text, email (with validation), select, and textarea fields all accept input correctly	Pass
TC-06	Form Buttons	Click Submit and Reset	Submit posts the form data; Reset clears all entered values	Pass

OUTPUT

Output 1: Pure Semantic HTML Static Webpage with Header, Articles, Sidebar and Contact Form



Thus, the pure semantic HTML5 static webpage demonstrating structural elements, table-based layout, list-based sidebar content, and a native contact form was successfully designed, implemented, and verified.